- Second harmonic ($n = 2$): $270 \times 2 = 540$ Hz.

18 C

$$Q = mc\Delta\theta$$

$$P = mc\frac{\Delta\theta}{t}$$

$$P = mc(\frac{4}{60})$$

$$Q = mL$$

$$P(40 \times 60) = mL$$

Hence,

$$c(\frac{4}{60}) = \frac{L}{40 \times 60}$$

$$\frac{c}{L} = \frac{1}{160} = 0.0063 \text{ K}^{-1}$$

19 D $1\ Nm$